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Math Methods-I (PHY421)

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Homework-5

1. Determine singularities, their nature and residues of the following functions:

$$\tan(z), \cot(z), \frac{\cot z}{z}, \frac{\csc z}{z^2}, \frac{\cot z}{z^2}, \frac{\csc z}{z^2}, \frac{\tanh z}{z^2}, \frac{z}{e^{z^2}-1}, \frac{e^{1/z}}{1-z},$$
$$\frac{1}{e^{1/z}+1}, \frac{z}{e^{z^2}-1}, \frac{1}{e^{pz}-1}, \frac{1}{e^{pz}+1}, \frac{e^{pz}}{\cosh^2 z}$$

with p is a real constant.

2. Using complex analysis, evaluate the following integrals:

$$\int_0^{2\pi} \frac{d\theta}{a \pm b \cos \theta}, \quad \int_0^{2\pi} \frac{d\theta}{a \pm b \sin \theta}, \quad \int_{-\infty}^{\infty} \frac{x^{2m}}{x^{2n} + a^{2n}} dx, \quad \int_{-\infty}^{\infty} \frac{x \sin(kx)}{x^2 + a^2} dx,$$

Here $a > b \geq 0$, $0 \leq m < n$ with m, n being integers, and $-\infty < k < \infty$.

3. Show that

$$\int_0^{2\pi} \delta(a - b \cos \theta) d\theta = \frac{2}{\sqrt{b^2 - a^2}}, \quad b > a$$
$$= 0, \quad b < a.$$

Hints: Use the Lorentzian representation of the Dirac delta function:

$$\delta(a - b \cos \theta) = \lim_{\epsilon \rightarrow 0} \frac{\epsilon/\pi}{(a - b \cos \theta)^2 + \epsilon^2}.$$

$$\int_0^{2\pi} \delta(a - b \cos \theta) d\theta = \lim_{\epsilon \rightarrow 0} \frac{\epsilon}{\pi} \int_0^{2\pi} \frac{d\theta}{(b \cos \theta)^2 - 2ab \cos \theta + a^2 + \epsilon^2}.$$

To apply residue theorem, we follow standard procedure: $\cos \theta = (z + 1/z)/2$.

4. Solve the following integral using (i) argument principle and (ii) residue theorem:

$$\oint_C \frac{f'(z)}{f(z)} dz,$$

for $f(z) = \cos(z)/z$ and C describes a circle of radius 2π centered at $z = 0$.

5. Using the Dirac identity, show that

$$\int_{-\infty}^{\infty} \frac{1}{(x \pm i\epsilon)} \cdot \frac{1}{(x^2 + 1)} dx = \mp i\pi$$
$$\int_{-\infty}^{\infty} \frac{x e^{2ix}}{x^2 - 1} dx = i\pi \cos(2).$$

6. (a) Show that the complex integral representation of the unit step function is

$$\theta(k) = \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x - i\epsilon} dx.$$

(b) Show that complex integral representation of Kronecker delta function is

$$\delta_{m,n} = \frac{1}{2\pi i} \oint_C z^{m-n-1} dz,$$

where m, n are integers and C is any circle centered at the origin.

7. Using the “indented” contour and “ $i\epsilon$ ” method, evaluate

(a)

$$\text{P.V.} \int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}. \quad (1)$$

(b)

$$\int_{-\infty}^\infty \frac{x \sin(kx)}{x^2 - a^2} dx$$

8. Using the Dirac identity (real-axis version of Sokhotski–Plemelj theorem), show that

(a)

$$\int_{-\infty}^\infty \frac{e^{ik_x x}}{k_x^2 - k_0^2} dk_x = -\frac{\pi}{k_0} \sin(k_0 x).$$

Note that the same result is obtained in class using two different methods: “indented” contour and “ $i\epsilon$ ” method.

(b)

$$\int_{-\infty}^\infty dE \int_0^\infty dt f(E) e^{-iEt} = \pi f(0) - i \text{P.V.} \int_{-\infty}^\infty \frac{f(E)}{E} dE.$$

(c)

$$\int_0^\infty e^{-i(\omega \pm \omega_0)t} dt = i \text{P.V.} \left[\frac{1}{\omega \pm \omega_0} \right] + \pi \delta(\omega \pm \omega_0).$$

(d)

$$\delta(x - x_0) = \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi i} \left[\frac{1}{x - x_0 - i\epsilon} - \frac{1}{x - x_0 + i\epsilon} \right].$$

9. Using the “ $i\epsilon$ ” method, show that

$$G(\mathbf{r}, \mathbf{r}') \equiv \int \frac{d^3 k'}{(2\pi)^3} \left[\frac{e^{i\mathbf{k}' \cdot (\mathbf{r} - \mathbf{r}')}}{k^2 - (k')^2} \right] = -\frac{1}{4\pi} \frac{e^{\pm i k |\mathbf{r} - \mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|}.$$

The function $G(\mathbf{r}, \mathbf{r}')$ is known as the Green’s function or the propagator for a free particle.